# AMR SENTINEL

**Antimicrobial Resistance Intelligence Platform**

**REPORT ID:** REP-5BCDEC0C
**DATE:** 08 Oct 2026 08:49 UTC
**STATUS:** Research Authorized

| | | | |
|---|---|---|---|
| **Pathogen Species:** | *Escherichia coli* | **Investigator:** | Dr. Sarah Chen, Ph.D. |
| **Isolate Identifier:** | ISO-202610-EC0C | **Institution:** | Institute of Antimicrobial Genomics |
| **Test Modality:** | Kirby-Bauer Agar Disk Diffusion | **Reference Standard:** | CLSI M100-ED33:2023 |
| **Calibration:** | known_disk_6mm (5.0 px/mm) | **Analysis Engine:** | CV Zone Radial Profiler v1.0 |

## EXECUTIVE PHENOTYPIC SUMMARY

| | | | |
|---|---|---|---|
| **Tested Antibiotics:** 3 | **Resistant (R):** 2 | **Intermediate (I):** 0 | **Susceptible (S):** 1 |

## QUANTITATIVE ZONE MEASUREMENTS & BREAKPOINT INTERPRETATION

| Antibiotic | Code | Disk (mm) | Zone (mm) | Result | CLSI Standard Breakpoint |
|---|---|---|---|---|---|
| **Ciprofloxacin** | CIP | 6.0 | **12.4 mm** | **R** | S: >= 26 mm \| R: <= 21 mm |
| **Gentamicin** | GEN | 6.0 | **21.0 mm** | **S** | S: >= 15 mm \| R: <= 12 mm |
| **Ampicillin** | AMP | 6.0 | **8.3 mm** | **R** | S: >= 17 mm \| R: <= 13 mm |

**METHODOLOGICAL LIMITATIONS & SURVEILLANCE BOUNDARIES:**

- This phenotypic assessment relies on automated Kirby-Bauer disk diffusion zone segmentation.
- Zone measurements are converted using standard 6.0 mm disk physical calibration. Micro-colonies or fuzzy margin zones should be verified manually.
- Phenotypic observations do not predict the future evolutionary trajectory of this specific isolate.

***LEGAL & SCIENTIFIC DISCLAIMER:*** *AMR Sentinel is an academic research and educational prototype. It is not cleared or approved by any medical regulatory agency for clinical diagnostic use, antimicrobial prescription selection, or direct patient therapeutic decisions. All outputs must be reviewed and confirmed by qualified clinical microbiology personnel.*